

HW 03

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Exercise 1

Tools and resources used:

Exercise 1 **Points attempted: 2.** **Time spent: 5 minutes.** **Difficulty: 1/10.** **Self-assessed score: 1.** **Questions/Feedback: None.**

Choice Exercise 2 **Points attempted: 2.** **Time spent: 5 minutes.** **Difficulty: 1/10.** **Self-assessed score: 2.** **Questions/Feedback: None.**

Choice Exercise 3 **Points attempted: 6.** **Time spent: 30 minutes.** **Difficulty: 3/10.** **Self-assessed score: 6.** **Questions/Feedback: Mathematical correctness, clarity.**

Exercise 2

Done! I discussed my solution to HW02 Exercise 9.4.

Exercise 3

1. ‘ ‘ Let $(G, *) = (\mathbb{Z}, +)$, and let $a = 1 \in G$ and $b = 3 \in G$. Compute $a^4 * b^{-2} \in \mathbb{Z}$.

We see that $a^4 * b^{-2}$ expands to $1 + 1 + 1 + 1 - 3 - 3 = -2$.

2. Let $g \in G$, and let $\langle g \rangle$ be the cyclic subgroup of G generated by g . Prove that the order of the element g is equal to the order of the subgroup $\langle g \rangle$.

Proof. First, we notice that the order of $\langle g \rangle$ is countably infinite. Thus, traditional counting techniques may not be used.

We can rewrite all elements of $\langle g \rangle$ as $\{g^n \mid n \in \mathbb{Z}\}$. We can also rewrite the subgroup generated by g as $\{g^n \mid n \in \mathbb{Z}\}$.

Therefore, these two group are exactly the same, and have exactly the same order. $\square$

3. Consider the cyclic group C_{20} , where r is an 18° rotation, and thus $|r| = 20$. Find all elements $r^n \in C_{20}$ such that $\langle r^n \rangle = C_{20}$.

For all $n \in \{1, 3, 7, 9, 11, 13, 17, 19\}$, $\langle r^n \rangle = C_{20}$.

4. Give an example of an infinite group $(G, *)$ and a subset $H \subseteq G$ which is closed under $*$, but where H is not a subgroup of G .

Let $H = 2\mathbb{Z} + 1$. H is closed under $*$ but does not include an identity element, so it is not a subgroup of G .

5. Prove that if $a \equiv a' \pmod{n}$, $b \equiv b' \pmod{n}$, and $a + b \equiv c \pmod{n}$, then

$$a' + b' \equiv c \pmod{n}.$$

Proof. By definition, $a - a' = kn$ and $b - b' = ln$. If $a + b \equiv c \pmod{n}$, then $a + b - c = jn$.

With some algebra, we find

$$a = kn + a', b = ln + b'$$

.

Then, we write $kn + a' + ln + b' - c = jn$. We can then subtract over some terms, then factor to get

$$a' + b' - c = jn - kn - ln$$

$$a' + b' - c = (j - k - l)n$$

.

Therefore, $a' + b' \equiv c \pmod{n}$. $\square$